

WHO INVENTED “FUNCTION”?

The word was introduced by Gottfried Wilhelm Leibniz in a letter of 1673, to describe quantities related to the points of a curve, such as a coordinate or the slope. The notation $f(x)$ appeared around 1734, with Alexis Clairaut and Leonhard Euler. For Euler, in his influential textbook of 1748, a function was an analytic expression built from the variable and constants – a single formula. Functions defined piece by piece, like the absolute value or the sign function above, do not fit this definition, and the concept gradually widened; Euler himself gave a broader definition in 1755, in terms of quantities that change when others change. The modern idea of a function as any rule assigning to each x a unique value, whether or not it is given by a formula, is traditionally credited to Nikolai Lobachevsky (1834) and Peter Gustav Lejeune Dirichlet (1837), independently. In the 20th century the definition was finally phrased entirely in terms of sets: a function is identified with its graph, a subset of the Cartesian product $X \times Y$.

4.2 Range and pre-image

Definitions

Let $f : X \rightarrow Y$ and let $A \subseteq X$. The **image of A under f** is the subset of Y

$$f(A) = \{f(x) \mid x \in A\} \subseteq \text{im}(f) \subseteq Y.$$

Let $y \in Y$. The **pre-image of y under f** is the subset of X

$$f^{-1}(y) = \{x \in X \mid f(x) = y\} \subseteq \text{dom}(f) \subseteq X.$$

Let $B \subseteq Y$. The **pre-image of B under f** is the subset of X

$$f^{-1}(B) = \{x \in X \mid f(x) \in B\} \subseteq \text{dom}(f) \subseteq X.$$

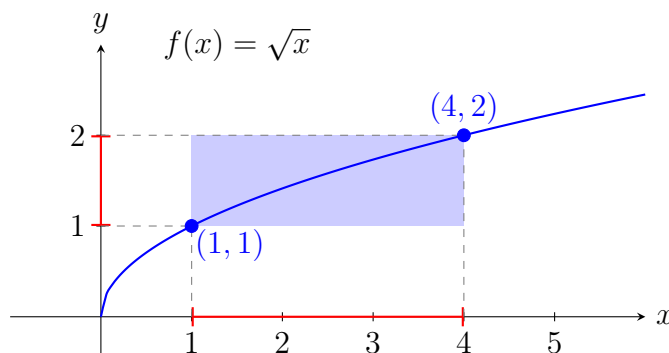
Notice that

- $f(X) = \text{im}(f)$.
- It is possible that $f^{-1}(y)$ or that $f^{-1}(B)$ are empty. For example, for $f(x) = x^2$, $f^{-1}(-5) = \emptyset$ and $f^{-1}([-4, -2]) = \emptyset$.

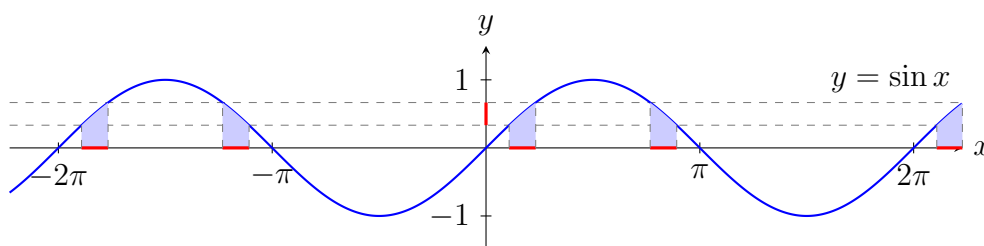
Example 4.2: Here are some examples of functions $\mathbb{R} \rightarrow \mathbb{R}$:

1. Let f be given by $f(x) = 2x$. Let $A = (a, b)$, where $a < b$. Then $f(A) = (2a, 2b)$. For any $y \in \mathbb{R}$, $f^{-1}(y) = \{\frac{y}{2}\}$. This is the *set* that contains the point $\frac{y}{2}$. Normally, in cases like this where it is clear that there's a single point within the set, we could write $f^{-1}(y) = \frac{y}{2}$ (which is the point itself, and not the set containing the point).
2. Let f be given by $f(x) = 4$. Then for any non-empty $A \subseteq \mathbb{R}$, $f(A) = \{4\}$. Moreover, $f^{-1}(4) = \mathbb{R}$, while $f^{-1}(y) = \emptyset$ for any $y \neq 4$.

3. Let $f(x) = \text{sign}(x)$. Then $f([0, 1]) = \{0, 1\}$, and $f^{-1}(-1) = \mathbb{R}_-$. Note that $f(0) = 0$, and $f(\{0\}) = \{0\}$.
4. Let $f(x) = \sqrt{x}$. Then $f((1, 4)) = (1, 2)$, $f^{-1}([1, 2]) = [1, 4]$, $f^{-1}(-1) = \emptyset$.



5. For $f(x) = \sin x$, we can see that $f^{-1}([\frac{1}{3}, \frac{2}{3}])$ is the union of infinitely many intervals.



We can now talk about the supremum, infimum, maximum and minimum of the image of various sets under a real-valued function f :

Supremum and infimum of a real-valued function

Let $f : X \rightarrow \mathbb{R}$ be a real-valued function. Let $A \subseteq \text{dom}(f)$. The **supremum of f on A** is the supremum of the image of A under f :

$$\sup_A f = \sup_{x \in A} f(x) = \sup\{f(x) \mid x \in A\}.$$

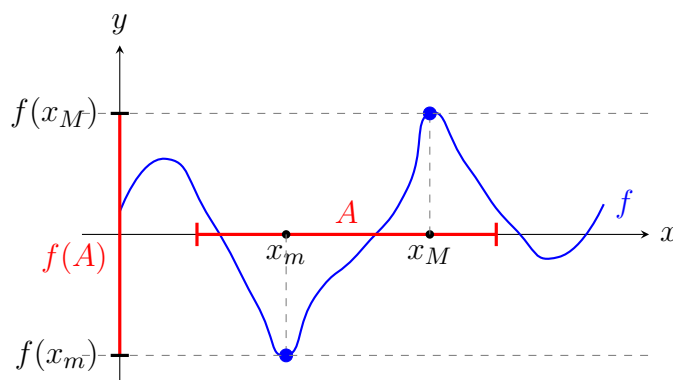
Similarly, the **infimum of f on A** is the infimum of the image of A under f :

$$\inf_A f = \inf_{x \in A} f(x) = \inf\{f(x) \mid x \in A\}.$$

As we have already seen, the supremum can be an element of $\mathbb{R} \cup \{+\infty\}$ and the infimum can be an element of $\{-\infty\} \cup \mathbb{R}$.

Boundedness of a real-valued function

If $\sup_{x \in A} f(x) < +\infty$ (i.e. it is a real number), we say that f is **bounded from above on A** . If $\inf_{x \in A} f(x) > -\infty$ (i.e. it is a real number), we say that f is **bounded from below on A** . If f is bounded from above and below on A , we say that it is **bounded on A** .



Maximum and minimum of a real-valued function

If $\sup_{x \in A} f(x) < +\infty$ and it belongs to $f(A)$ then it is the **maximum of f on A** . It is denoted

$$\max_A f \quad \text{or} \quad \max_{x \in A} f(x).$$

If $\inf_{x \in A} f(x) > -\infty$ and it belongs to $f(A)$ then it is the **minimum of f on A** . It is denoted

$$\min_A f \quad \text{or} \quad \min_{x \in A} f(x).$$

Since the minimum and the maximum of f on A belong to $f(A)$, there exist $x_m \in A$ and $x_M \in A$ such that

$$f(x_M) = \max_A f \quad \text{and} \quad f(x_m) = \min_A f.$$

Example 4.3: 1. For $\sin x$, we have

$$\max_{x \in \mathbb{R}} (\sin x) = 1 \quad \text{and} \quad \min_{x \in \mathbb{R}} (\sin x) = -1.$$

2. For x^2 we have

$$\sup_{x \in \mathbb{R}} (x^2) = +\infty \quad \text{and} \quad \min_{x \in \mathbb{R}} (x^2) = 0.$$

If $A = [-10, -3]$ then

$$\max_{x \in A} (x^2) = 100 \quad \text{and} \quad \inf_{x \in A} (x^2) = 9.$$

Note that in this last case, the infimum is not achieved, so there's no minimum on A .

4.3 Surjectivity, injectivity, and invertibility

Let us define some important *global* properties of functions, i.e. properties that tell us something about the function as a whole.